

YOLO Objective: Object detection

You look only Once

Input: - Image

Output: - Bounding box Parameters

Prediction = $\hat{Y}$ Ground $T = Y$

$\begin{bmatrix} p_{obj} \\ x \\ y \\ w \\ h \\ p_{class} \end{bmatrix}$

category of
object in
the image

Approach in simple Models

* limitation? what if we have
more than one class!

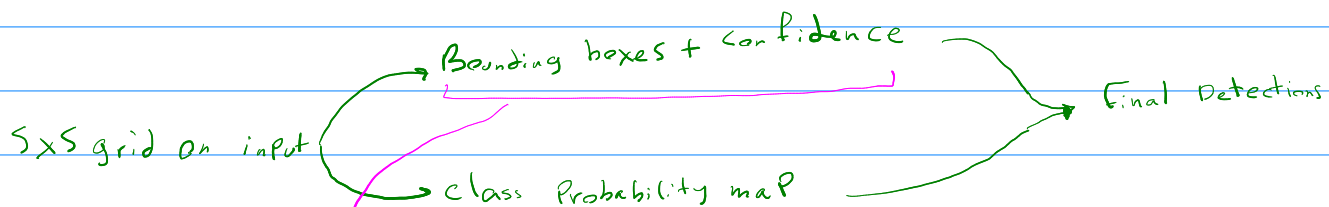
we need to focus on: ① Input

② Neural Network Architecture

③ Output

④ Ground Truth

⑤ Loss

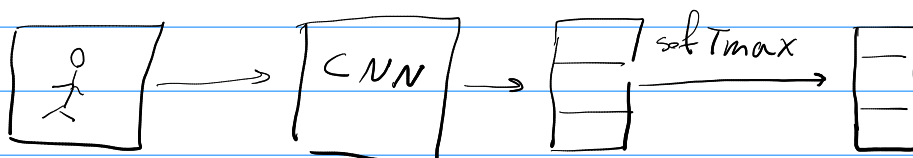


$[p_{class 1}, \dots, p_{class n}]$

* Each class has
'n' class probabilities
and 'B' bounding boxes

why multiple
bounding boxes per grid
cell? so that we can
predict more than one
object.

$\begin{bmatrix} x_1, y_1, \sqrt{w_1}, \sqrt{h_1}, C_1 \\ x_2, y_2, \sqrt{w_2}, \sqrt{h_2}, C_2 \\ \vdots \\ x_B, y_B, \sqrt{w_B}, \sqrt{h_B}, C_B \end{bmatrix}$



we normalize bounding boxes to get $0 \leq x, y \leq 1$ & $0 \leq w, h \leq 1$

$\begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \\ 0.7 \\ 0.3 \\ 0.6 \end{bmatrix} \rightarrow \begin{matrix} p_{obj} \\ x \\ y \\ \sqrt{w} \\ \sqrt{h} \\ c \end{matrix}$

So all will be normalized.

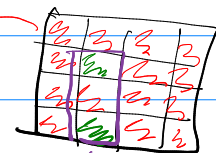
$\downarrow \downarrow$
 $\frac{x \cdot 1.5}{s} \quad \frac{y \cdot 1.5}{s}$
 $\frac{w}{w} \quad \frac{h}{h}$

for $s \times s$ grid on input

How do we get confidence score? Center of each class will have confidence score of 1 and 0 for rest

$$C = \text{IoU}(P_{pred}, b_{true}) \times P(\text{object} = 0)$$

$$C = \text{IoU} \times P(\text{object}) = 1$$



read it as

$(x, y, \sqrt{w}, \sqrt{h}, c) \Rightarrow (0.2, 0.4, 0.8, 0.6, 0.9)$
 $(P(\text{dog}), P(\text{bike}), P(\text{cat})) \Rightarrow (0.8, 0.1, 0.1)$

I am 90% confident that an object exists in the current grid cell box with $\sqrt{w} = 0.8, \sqrt{h} = 0.6$ and the location of bounding box is $x = 0.2, y = 0.4$ and that object is 80% likely a dog.

$P(\text{obj}) = 0.9$
 $P(\text{class}) = 0.8, 0.1, 0.1$
 $0.9 \times 0.8 = 0.72, 0.09, 0.09$

* we need to Parallelize for ① wrong class ② Bounding box

③ Confidence

what does the loss looks like? During inference we trust the model's ability to approximate $\text{IoU} \times P(\text{object}) = C$. The model outputs C directly for each bounding box. It does not compute IoU.

C is the score for any object being present, class wise score will be $C \times P(\text{class})$. So err

$$\text{Loss} = \underbrace{L_{\text{class}}}_{\text{Class Prob}} + \underbrace{L_{\text{localization}}}_{\text{Bounding box + Confidence}}$$

$$L_{\text{CLASS}} = \sum_{i=1}^C (p_{ci} - \hat{p}_{ci})^2$$

only for grid cells with centroids

$$L_{\text{coordinates}} = L_{\text{confidence}} + L_{\text{coordinates}}$$

$$L_{\text{coordinates}} = \lambda_{\text{word}} \times [(x - \hat{x})^2 + (y - \hat{y})^2 + (w - \hat{w})^2 + (h - \hat{h})^2]$$

only for grid cells with centroids

$$L_{\text{confidence}} = \underbrace{\sum (c_i - \hat{c}_i)^2}_{\text{for grid cells with centroids}} + \underbrace{\lambda_{\text{no-obj}}}_{\substack{\text{small} \\ \text{number} \\ \text{between} \\ 0 \text{ and } 1}} \underbrace{\sum (c_j - \hat{c}_j)^2}_{\text{for grid cells w/o centroids}}$$

What are Yolo limitations?

① Each cell only contain one class

② Each grid cell can contain only '1' object of 1 class
- Problem in crowded images.

③ Unusual aspect ratio orientation of the same object is problem

④ Impact of error in small boxes are higher